

Institut für Linguistik
Heinrich-Heine-Universität Düsseldorf
Dr. Yulia Zinova
Summer 2026



Computational Morphology

Māori via Xfst - Documentation

Marie Vangerow-Kühn (3694419)
October 5, 2026

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1 - Intro

Te reo Māori, the language of the indigenous Māori people of New Zealand, belongs to the Eastern Polynesian languages such as Hawaiian and Tahitian. A reduced Latin alphabet (see '2 - Validity') was enforced in the late 18th century by native English speakers.

1.1 - Objective

This paper summarizes the development process in preparation for the oral exam. As a result of an alignment with peers, this work focuses on verbal morphology (including the adjacent word classes universals and statives) to demonstrate Xfst (Xerox finite-state tool) functionality and to show broad coverage of high-frequency vocabulary.

Although Māori morphology is less inflectional than in certain Indo-European languages, there are distinct phenomena suitable for demonstrating Xfst's features.

1.2 - Xfst Features

Xfst transforms formal descriptions into finite-state morphology. In short, according to Beesley and Karttunen (2003: p.ix-x) this tackles two main morphological challenges:

1. **Morphosyntax:** Words are composed of morphemes, bounded by constraints via finite-state networks.
2. *Alternation:* Variations in form like fly/flies are handled by transducers that produce systematic shifts.

Both are composed into a single **lexical transducer**, which allows bidirectional operations on large vocabularies with few resources.

1.3 - Scope

The model maps surface forms (lower words) like *whakarunga* to their corresponding abstract lexical forms (upper words), in this example *caus+runga+Loc*. Morphological tags follow the abbreviations established by Bauer et al. (1993: p. xxvii ff.) with adaptations such as Arabic numbers instead of the Roman ones and capitalization for readability (see section 'Tags').

Rules are numbered in both main files (*maori.xfst* and *maori-lexc.xfst*) and listed in 'Rules'. These files contain plain text; the *.xfst* extension is chosen to enable syntax highlighting.¹

2 - Validity

Besides referencing the lexicon (*maori-lexc.xfst*), the main file (*maori.xfst*) contains validity restrictions, like the **restricted alphabet**, which reflects that Māori uses fewer letters:

- 8 consonants (h, k, m, n, p, r, t, w)
- 2 digraphs (ng, wh)
- 5 vowels (a, e, i, o, u),
each with a long version, denoted via macrons (ā, ē, ī, ō, ū)²

In addition, there is a **restricted syllable structure**: "the *consonants* always stand singly, and every syllable ends with a vowel" (Harawira, 1974: p. 12). As a result, only the combinations V, VV, CV and

¹Based on https://www.vim.org/scripts/script.php?script_id=3441.

²Formerly not separated, partially via vowel doubling.

CVV are allowed in Māori, equal to (C)V(V). Based on these restrictions, ungrammatical strings are excluded to avoid overgeneration. These rules were also used for the Māori guesser (see '5 - Quality Assurance and Testing').

During testing (see '5 - Quality Assurance and Testing'), the need for certain rewriting rules was identified; e.g. whaka+āe results in whakaāe.

3 - Lexicon / Word Forms

The lexicon file (maori-lexc.xfst) is organized according to word classes. Māori words are classified following Harlow (2007: p. 97 f.)³ with five distinct base classes of words, accompanied by functional particles:

- **bases:**
 1. (common) nouns
 2. verbs or universals which allow passive usage
 3. statives including neuter verbs and adjectives
 4. locatives with location words and locative nouns
 5. personals including personal nouns and names as well as pronouns and numerals
- **minor morphemes:** "All non-base morphemes" (Biggs, 1961: p. 18), particles including determiners and interjections

As stated by Boyce (2006: p. 52), there might have been colonial influence: "Early manuscripts written in Māori by Māori show a high degree of variance in the placement of word boundaries, and this may be an indication of the need to develop a new awareness of 'word' as a concept when writing."

Since many **bases** function both nominally and verbally, the first lexicon contains such stems, leading to a continuation lexicon which branches into both +N and +V tags. One insight from a later stage of this project is that most verbs have the potential to become nouns (or, more generally: most stems do have the potential to function across multiple Western lexical categories). Therefore, the label 'NandV' initially introduced for bases, was also used within other lexica (see '3.2 Verbs'). Later, 'VandAdj' was introduced for tagging participial adjectives.

Characteristically, Māori lexical items have many functions and meanings. For example, "*mea*" can be used, following Harawira (1974: p. 51) and the TeAka Dictionary.

- as a noun: *thing, object, property, reason, thingy* - and as a loanword: *mayor*,
- as a verb: *make, do, deal with, cause, say, intend, think*,
- as particle: *soon (in time)*,
- in the combination "he mea" for past tense and as an alternative for a passive suffix,
- in "nō/nā te mea" meaning *because*, and
- as a placeholder for "such-and-such, so-and-so, somewhere, at some time" - when a name/location, ...is not specified.

³Referencing Biggs (1961): six base classes, called N, V, A, G, L and P with G as a "class of very wide distribution substituting for N, V, and A bases" (p. 25).

Consequently, the tagging as well as the translations contained in the lexicon file should be read as examples. Full coverage is beyond the scope of this work.

Overall, the following lexicon structure was created (simplified representation):

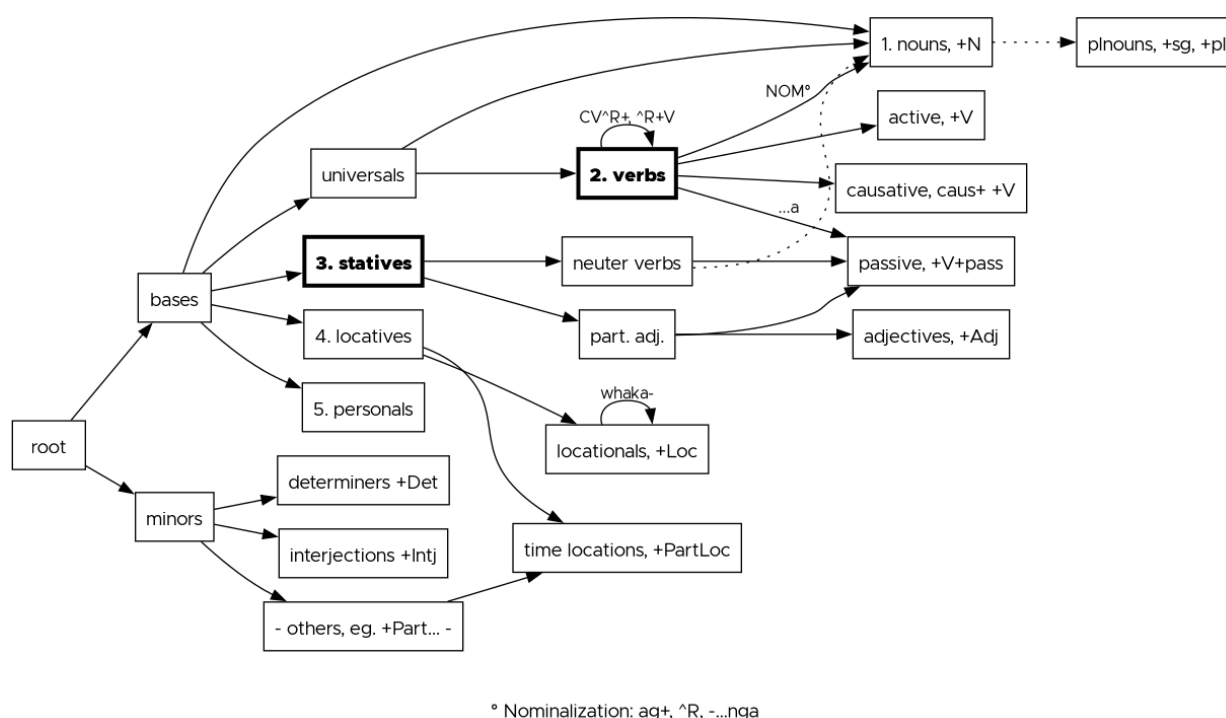


Figure 1: Structural representation of the contents of the main files

3.1 Common Nouns

Since they are not the focus, nouns are simply listed in flat lexica. Distinctions are made for nouns, which exceptionally do have plural forms (Maunsell, 1862: p. 21) and for calendar terms. A label like +ca1 was considered but dismissed, to avoid mixing up structure and meaning.

Even though they are not explicitly labeled, the loanwords from English are worth mentioning: (Harawira, 1974: p. 13): *hōiho* (horse), *kau* (cow), *hu* (shoe), *pepa* (paper), *pata* (butter) or *huka* (sugar).

3.2 Verbs

Verbs are the central and most extensive part of this work. According to Aubert (1885: p. 11): "They may be classified into active, passive, and neuter, the passive form being very extensively used, more so perhaps than in any other known language." Māori verbs lack inflection for person, number or genus (Maunsell, 1862: p. 35). Tense, aspect and mood are realized via particles (see '3.6 - Particles')

Certain verbs are listed in the lexicon file as stems, even if they appear to be composed of prefixes, such as *whakatō* (to sow, insert). They were added because they appeared within lists of frequent words (see '5 - Quality Assurance and Testing') but not as a composition, if the stem could not be found in Moorfield (2026).

Prefixes

Above the split between basic word classes, prefixes are modeled as branches directly from the root, according to Harawira (1974: p. 50):

- *whaka-*, which "creates transitive verbs from intransitives, including neuter verbs and adjectives" (Bauer et al., 1993: p. 515)⁴:

tupu (to grow, increase) ⇒ **whaka**tupu (cause to grow), caus+**tupu**+V

- *tā-* as a causative prefix, which also applies to nouns and adjectives:

mahana (be warm) ⇒ **tā**mahana (warm up, reheat food), caus+mahana+V

- *kai-* functions as an agentive prefix, creating a noun used for an acting human:

peita (to paint, (loan)) ⇒ **kai**peita (painter), ag+peita+N

Those prefixes are realized with **flag diacritics** (aka 'labels') like @P.PREF.Kai@ (= **P**ositive set, the prefix-feature **PREF** is to be set to the value **Kai**). This allows adding the individual prefix (and its label for upper words) at the beginning, in contrast to the usual left-to-right order in Xfst.

-nga Nominalization

Another process for nominalization is also called "-Canga nominalization" (Bauer et al., 1993: p. 512). It is characterized by the variety of endings: *-nga*, *-anga*, *-inga*, *-hanga*, *-ranga*, *-kanga*, *-tanga* and *-manga*. For example:

noho (sit, stay) plus -anga ⇒ noho**anga** (place for sitting, chair), noho+nom+N

Usually one verb stem allows only one of the different suffixes. Therefore they are realized as separate continuation classes.

Passive

There are several passive suffixes. Besides the *-a* ending, individual continuation classes are modeled for *-na*, *-nga*, *-ia*, *-tia*, *-hia*, *-ngia*, *-kia*, *-mia*, *-ria*, *-ina*, *-whia*, *-whina*, *-hina*, *-kina* - and also without a special suffix (0Passiv), which occurs especially in classical Māori. This results in e.g.:

mahi (to work) ⇒ mahi**a** (to be worked on/through), mahi+Pass+V

hopu (to catch, seize) ⇒ hopu**kina** (to be caught/seized), hopu+Pass+V

This approach was chosen over flag diacritics for clarity and usability. This also allows simply adding -nga nominalizations where applicable and could be used as extra continuation classes in other lexica

⁴Also describing whaka-prefixation for nouns, turning them into verbs.

as well. For each passive form allowed for a verb, one line in LEXICON Verb is added (which could be several lines). As an example:

```
kāwana  VNtanga ;
kāwana  tiaPassive ;
```

This results in

- kāwana (to govern), kāwana+V
- kāwanatanga (government, rule, authority), kāwana+Nom+N
- kāwanatia (being governed), kāwana+Pass+V

Reduplication

There are different reduplication patterns affecting Māori verbs. **Initial partial reduplication** (CV ⇒ CVCV) expresses repetition and intensifies a verb. This is a form of fixed-length reduplication, comparable to the course example from Tagalog (Beesley and Karttunen, 2003: p. 479 f.). In Xfst, this is realized via rewrite rules (paraphrased: *"initially add consonant and add vowel for marked verbs"*), and applied to the verb class in LEXICON IniDupVerbs, where ConVow is added as a marker in the following continuation class. This results in e.g.:

kata (laugh) ⇒ kakata (frequent laughing), CV[~]R+kata+V
 tiro (look at, inspect) ⇒ titiro (to observe, survey), CV[~]R+tiro+V

Full-stem reduplication occurs with a similar meaning and also for nominalization:

tā (strike, beat) ⇒ tātā (to strike repeatedly), tātā[~]R+V
 kawē (to carry) ⇒ kawekawē (freight and cartage). kawē[~]R+N

This was modeled using brackets in the lexicon file and enforced via `compile-replace lower ;` in the main file. Thus, lower words were transformed. For example, [~][{kau}[~]2[~]] resulted in *kaukau*.

For **final partial reduplication** (mākū ⇒ mākūkū, pūtai ⇒ pūtaitali) there seem to be very few examples, but with variations: stemCV ⇒ stemCVCV for diphthongs, and stemConV1V2 ⇒ stemCV1V2V1V2 for two vowels. As of runtime, this feature is not included in the main lexc-file (see '4 - Incremental Approach'), but is available as a building block.⁵

There are more reduplication patterns in Māori (see e.g. Bauer et al. (1993: p. 525 ff.)), but they appear to be less frequent.

3.3 Statives

In English there is a category of stative verbs that do not typically occur in the progressive tense. In Māori there are words that can be used as verbs, but not with passive endings.⁶ In general, such

⁵Files: Partial-FinDup.xfst and Partial-FinDup-lexc.xfst

⁶"A STATIVE is any base which can be used verbally but not passively." (Harlow, 2007: p. 97)

grammatical states or state-changes do not function as nouns. But, as often in Māori, there are exceptions.

This category contains **neuter verbs** (+V+pass), which predicate states or conditions (e.g. *mate*, "be sick, dead")⁷ and **adjectives** (+Adj) - including colors, which are listed in a separate lexicon. Participial adjectives can function as attributive adjectives or passive verbs (Maunsell, 1862: p. 54). Accordingly, they are tagged twice via a continuation class, branching into two labels:

ea (be satisfied, be (re)paid) ⇒ *ea*+Adj, *ea*+Pass+V
wera (hot, burnt) ⇒ *wera*+Adj, *wera*+Pass+V.

Reduplication is also used for adjectives: A slight decrease in quality similar to the -ish suffix in English is produced via doubling an adjective's root. Ngata (1979: p. 10 f.) provides two examples: *wera* (hot, burnt) ⇒ *wera**wera* (warm), *pango* (black) ⇒ *pango**pango* (blackish). As this falls outside the scope, it is not modeled.

3.4 Locatives

This class includes locations⁸ and some other words "related to position in space or time" (Biggs, 1961: p. 26). Those are tagged with +Loc. Locations and locative nouns usually allow the *whaka*-prefix as a directional prefix (cause to move toward ...). Exceptions are listed in this lexicon using the disallow flag diacritic: @D.PREF.Whaka@ (= Disallow, the prefix feature **PREF** does not allow the value **Whaka**). Examples include:

waho (the outside, the open sea, offshore), *waho*+Loc
*whaka**waho* (outwards, seawards), *caus*+*waho*+Loc
tai (the sea, coast), *tai*+Loc

3.5 Personals

Besides pronouns, the word class of personals contains numerals and proper names (people, tribes, stars, canoes), which are not reflected here. Personal, possessive and interrogative pronouns are listed with translations and tagged based on Ngata (1979: p. 14 f). Derivation is not modeled, as not in focus.

Pronouns

As in other Polynesian languages, the **personal pronouns** (+Pers) are distinguished by singular, dual, and plural, as well as between inclusive and exclusive in the first person. As a result, there are, for example, different interpretations of the English concept "we":

⁷"The neuter expresses being, or a state or condition of being; when the agent and the object acted upon coincide, and the event is properly neither action nor passion, but rather something between both: as *I am, I sleep, I walk.*" (Maunsell, 1862: p. 44)

⁸Place names such as mountains rivers, or regions etc. are not included here.

- *tāua* (= you + me, +1d1-incl,
- *māua* (= me + another one, +1d1-excl),
- *tātou* (= we/us, 3+, includes listener, +1p1-incl),
- *mātou* (= we/us, 3+, excludes listener, +1p1-excl).

The **possessive pronouns** (+Poss) also distinguish between different classes of relationships (Bauer et al., 1993: p. 210):

- the *ā*-class, where a possessor dominates (*nāku* (mine / by me)), and
- the non-dominant *ō*-class: *nōku* (mine / belonging to me).

Others, including interrogatives like *wai* ("who?"), are tagged as pronouns (+Pron), interrogatives receiving an additional +q tag.

Numerals

For the distinction between cardinal, ordinal and personal numbers for the digits 1 to 9, the LEXICON Numerals starts by defining specific prefixes, resulting in e.g.:

rua (two, both), card+rua+Num
tuarua (second), ord+rua+Num
tokorua (two people, couple), pers+rua+Num

Larger numbers, formed by connecting terms with the particle *ma* (and), are not represented. Some examples from Ngata (1979: p. 11):

- *tekau ma tahi* (11) is a combination of *tekau* (10) and *tahi* (1),
- *rua tekau* (20) consists of *rua* (2) and *tekau* (10),
- *kotahi rau*: 100, and
- *kotahi mano*: 1,000.

3.6 - Particles

Harlow (2007: p. 97) stated: "Particles are those words which occupy the peripheries of phrases and fulfill such functions as case marking, tense/aspect marking, directionality, deixis, and so on." Given their high frequency, they are listed in different lexica with tags and translations even if there are no derivational processes described:

- Determiners +Det, containing articles,
- Particles +Part, especially: manner +PartMnr, direction +PartDir and location +PartLoc,
- Tense, aspect and mood markers: +Tns, +Asp, +Mde
- Prepositions +Prep,
- Interjections +Intj, including negations +Neg

Interjections serve as a cute, short form for phrases, like *aua*, meaning "I don't know, don't ask", which also represents a negation (do not, don't).

4 - Incremental Approach

After an initial setup which covered the word classes and some basic features, an incremental approach was applied. Each new non-trivial feature was first set up and tested solely as a "building block", using dedicated files, e.g. `whaka-Prefix.xfst`. It was then integrated into the main files (`maori.xfst`, `maori-lexc.xfst`).

Not included in the main files are the following features:

- Final partial reduplication: `mākū` $\Rightarrow$ `mākūkū`, `mātai` $\Rightarrow$ `mātaimai` (`Partial-FinDup.xfst` & `Partial-FinDup-lexc.xfst`), as it is not a primary focus and is less productive, combined with having a considerably long runtime due to >5,000 arcs.
- The historic *ma*-prefix, which builds statives in Polynesian languages, is not yet productive in Māori.

In cases of doubt as well as for contradictory sources, the Te Aka Māori online dictionary Moorfield (2026) was considered the gold standard due to the involvement of native speakers. For each term some (by no means all) English translations are added in the `lexc`-file to not stay meaningless.

As a starting point, a list of 205 Māori terms collected during initial research (`maori-words.txt`) supported preliminary checks. Later, the top 200 words in Boyce (2006: p. 266) was used to ensure adequate vocabulary coverage. Extra words were checked against Moorfield (2026) and, if available there, incorporated into the `maori-lexc.xfst`. As a counterpart, a list of malformed expressions (`maori-bad.txt`) was maintained.

As the lexicon evolved over time and functionality was expanded, some duplicates emerged. For example, when adding the LEXICON KaiVerbs, the disallow-tag for the *kai*-prefix caused an additional +V:

```
mahi+N
mahi+pass+V
mahi+V
mahi+V
```

5 - Quality Assurance and Testing

By running `print upper-words | sort > .../upper-words.txt`, an output file was generated and checked for such duplicates via Python, see `maori-xfst-tools.ipynb`⁹, excluding multiple passive tags per stem. Such issues were resolved manually.

The test cases in `maori-bad.txt` were correctly rejected, each returning "???". Valid words, including a longer compound, *tūrangawaewae* ("homeland", "place where one has the right to stand"), were accepted as expected.

In the later stages, the 20,000 most recent words from the Sketch Engine corpus, fetched through an API,¹⁰ were used for quality checks.

⁹Python scripts, built on the notebook `run_xfst_from_python.ipynb` provided by Sabine Hermes within the course.

¹⁰<https://www.sketchengine.eu/apidoc/openapi.yaml>

A check for over-generation delivered a delta of 98 generated lower words not present in the 20,000s. Notably, this included reduplications (e.g. *haupapa*, *ianaiane*, *kimokimo*, *matikuku*, *tāuwhiuwhi*) and passive forms (e.g. *horongia*, *hahauria*, *kohetitia*, *tanutia*). It is plausible that such terms are valid but infrequent.

The tests performed on September 18, 2026, combined with the first guesser run, yielded some suggestions for final extensions to the main files. In addition, some findings on the corpus emerged: First, it included English vocabulary (e.g. *dispersion*, *lab*, *and*, *new*, *brown*, *cook*, *jazz*) and words with a glottal stop (e.g. *ho'okahi*, *ha'i*, *mo'ui*, *kaikua'ana*)¹¹. This might be an indication of the corpus' quality (contamination, noise). Furthermore, there were duplicates contained: Removing them resulted in just 13,945 lines.

These tests also included stressing the guesser (*maori-guesser.xfst*)¹² with the 20,000-word baseline. Valid words (see '2 - Validity') were accepted (returned as the output of 'apply up'). It resulted in a delta of 3,751 (18.8 %) words classified as invalid. Besides the QS findings on the corpus mentioned above, compound terms (e.g. *tai-tokerau*, *whanga-nui-a-tara*, *paenga-whāwhā*, *ā-hāpori*) were also included. These, particularly terms starting with the particle *ā*-¹³, would be a reasonable extension for the guesser.

Within the unknown words, several passive endings were present but not followed up and only isolated dictionary extensions were made.

6 - Final Tests and Remarks

Final tests, run on October 3rd, used a de-duplicated version of the most frequent words from Sketch Engine, containing only well-structured terms (the delta from the checks above): 10,194 words¹⁴.

Running those via *apply up* through tagging yielded 9,538 lines of 'unknowns' (indicated by "???"), which corresponds to a low detection rate of 6.4 %. This is unsurprising, because the lower-word list at that point contained 995 entries. From the 1,000 most frequent words, 617 remained undetected. The detection was 38.3 %.

Overall, the analysis yielded mixed results, suggesting room for further investigation and enhancements. But overall, the project covered core features of Māori verbal morphology. Although these were identified within the defined scope, they were only marginally generalizable. Nevertheless, the functionality of the Xfst tool was showcased.

(3,202 words)

¹¹That is reminiscent of the Hawaiian 'Okina, which is usually not written in Māori. Both characters ' and ' occurred.

¹²Based on the *monish-guesser.xfst* developed during the course

¹³Creates a modifier, e.g. *waha* (mouth, voice), ⇒ *ā-waha* (oral, verbal, spoken) (<https://maoridictionary.co.nz/word/45>)

¹⁴This basis for comparison (*maori-20k-clean.txt*) is a compromise, as e.g. the compound words mentioned above are excluded. As a result, the figures are somewhat optimistic. However, on balance, this seemed to be the less unfair measure.

Rules

Numbering starts top-down with the initial setup from the alignment. Newer rules up from (40) are added chronologically. Dismissed rules are struck through for consistent numbering.

0. Validity & Bases

- (1) restrict alphabet
- (2) restrict syllable structure
- (3) LEXICON Root: prefixes and word forms like Nouns, Verbs etc. as basic branches
- (4) LEXICON Bases: terms that are used as nouns and as verbs
- (5) LEXICON NandV: split towards noun/verb tagging

1. Nouns

- (6) LEXICON Nouns: separation between different noun forms (Noun; PINouns; DateTime)
- (7) LEXICON Noun: tagging for nouns -> NTag
- (8) LEXICON NTag: +Noun
- (9) LEXICON PINouns: nouns -> SgNounTag, PINounTag
- (10) LEXICON SgNounTag: +Noun+Sg
- (11) LEXICON PINounTag: +Noun+Pl
- (12) LEXICON DateTime: months, weekdays

2. Verbs

- (13) LEXICON Verbs: -> Verb; VerbStems; VerbDupNomin;
- (14) LEXICON Verb: verbs -> VTag, VNnga, VNanga, VNinga, VNhanga, VNranga
- (15) LEXICON VNnga: -> VTag, NTag (nominalization)
- (16) LEXICON VNanga: -> VTag, NTag (nominalization)
- (17) LEXICON VNinga: -> VTag, NTag (nominalization)
- (18) LEXICON VNhanga: -> VTag, NTag (nominalization)
- (19) LEXICON VNranga: -> VTag, NTag (nominalization)
- (20) LEXICON VerbStems verbs -> CloseVBrack (full-stem reduplication)
- (21) LEXICON CloseVBrack: -> VTag plus closing brackets (ordinary & reduplicated form)
- (22) LEXICON VerbDupNomin: verbs -> CloseNBrack (reduplication verb => noun)
- (23) LEXICON CloseNBrack: -> VTag; NTag (verb vs. reduplication: verb => noun)
- (24) LEXICON VTag: +V

3. Statives, Personals, Numerals and Interjections

- (25) LEXICON Statives:¹⁵ -> NeuterVerbs; ParticipialAdjectives; Adjectives; Colours
- (26) LEXICON STag: tagging +V+pass ¹⁶
- (27) LEXICON Numerals Numbers:¹⁷ numbers 0 to 9 -> NumTag
- (28) LEXICON NumTag: +Num
- (29) LEXICON Particles: -> Dets; PartMnr; PartDir; PartLoc; Parts; Possessives; Dets;
- (30) LEXICON Dets: determiners incl. tagging, e.g. +Det+def+sg
- (31) LEXICON PartMnr: particles of manner, +PartMnr (manner)
- (32) LEXICON PartDir: particles of direction, **terms**, +PartDir (direction)
- (33) LEXICON PartLoc: particles of location, +PartLoc (location)
- (34) LEXICON Parts: other particles -> PartTag (others)
- (35) LEXICON PartTag: +Part
- (36) LEXICON Possessive pronouns:¹⁸ possessive pronouns incl. tagging, e.g. +Poss+3sg
- (37) LEXICON Interjections: interjections -> IntjTag and negations -> NegTag

¹⁵changed into branching

¹⁶A dedicated tag like +stativ was dismissed.

¹⁷name changed

¹⁸changed name, excluded determiners

4. Locatives

(38) LEXICON Locationals:¹⁹: positions in space -> LocTag

(39) LEXICON LocTag: +Loc

+ Additional Rules

(40) LEXICON aPassive +pass (-a suffix)²⁰

(41) LEXICON IntjTag: tagging +Intj

(42) LEXICON NegTag: tagging +Neg

(43) LEXICON Prefixes: dispatching for whaka-/tā- and kai-prefix via labels

(44) LEXICON WhakaVerbs: contains verbs that take the whaka-prefix

(45) Rewrite rule: aāe => aae (whakaāe => whakaae)

(46) LEXICON KaiVerbs: action verbs allow nominalization via the kai-prefix

(47) LEXICON KaiAgentiv: verb (disallow kai-) vs. noun (require kai-)

(48) LEXICON IniVDup: starting initial reduplication CVstem => CVCVstem

(49) LEXICON IniDupVerbs: contains verbs that allow initial partial reduplication

(50) CopyCon & CopyVow: rewrite rules for initial partial reduplication

(51) LEXICON Adjectives

(52) LEXICON Colours

(53) LEXICON AdjTag: tagging adjectives with +Adj

(54) LEXICON Numerals: cardinal-only numbers and prefixes (card+, ord+, pers+) for (27)

(55) LEXICON Prepositions

(56) LEXICON PrepTag for tagging prepositions with +Prep

(57) Rewrite rule: aa=>ā (e.g. for passive suffix -a)

(58) LEXICON VNkanga: kanga-nominalization (as a supplement for 14 ff.)

(59) LEXICON VNtanga: tanga-nominalization

(60) LEXICON VNmanga: manga-nominalization

(61) LEXICON NeuterVerbs: statives predicating a state.

(62) LEXICON ParticipialAdjectives: modifying nouns -> VandAdj

(63) LEXICON VandAdjv: branching for participial adjectives -> STag; AdjTag ;

(64) LEXICON Locatives: branching -> Locationals; TimeLocations

(65) LEXICON TimeLocations: positions in time, usually disallow whaka-prefix -> PartLocTag

(66) LEXICON Personals: branching to personal/possessive pronouns, numerals and others

(67) LEXICON PersPronouns: personal pronouns incl. tagging e.g. +Pers+1d1-incl

(68) LEXICON OtherPersonals: interrogative pronouns (+Pron+q)

(69) LEXICON PartLocTag: tagging for location Particles: +PartLoc

(70) LEXICON TnsAspMde: markers incl tags: +Tns +Mde +Asp

¹⁹previously Locatives, renamed with separating TimeLocations, see (64) & (65)²⁰additional rules for other suffixes were added later, see 'Passive'

Tags

^R	Reduplication
+1dl	first person dual
+1pl	first person plural
+1sg	first person singular
+2pl	second person plural
+2sg	second person singular
+3pl	third person plural
+3sg	third person singular
+Adj	adjective
+adv	adverbial
+afo	aforementioned
+Asp	aspect marker
+card	cardinal
+def	definite
+Det	determiner
+dl	dual
+excl	exclusive
+incl	inclusive
+indef	indefinite
+Intj	interjection
+Loc	locatives
+Mde	mood marker
+N	noun
+Neg	negation particle
+nom	nominalizing suffix
+Num	numeral particle
+ord	ordinal
+Part	(other) particle
+PartDir	direction particle
+PartLoc	location particle
+PartMnr	manner particle
+pass	passive
+pl	plural
+Pers	personal pronoun
+Poss	possessive pronoun
+Prep	preposition
+Pron	(other) pronoun
+prox1	near speaker
+prox2	near hearer
+q	question particle
+sg	singular
+Tns	tense marker
+V	verb
caus+	causative prefix

Setup Notes

- Xfst Software libxfs is 2.25.11. (<http://www.fsmbook.com> - via WaybackMachine, <https://web.archive.org/web/20220303082440/http://web.stanford.edu/~laurik/.book2software/>), installed in Kubuntu 26.04
- code written in Kate editor, syntax highlighting adapted from https://www.vim.org/scripts/script.php?script_id=344 (adapted)
- sources available in GitHub: <https://github.com/marievk/maori-xfst>

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Declaration of AI Usage

I hereby confirm that I have read and acknowledged the AI policy of the Institute for Linguistics, HHU (June 2025). I hereby declare that for this project I have used AI. The following list contains all tools that I have used as well as a short description of how and for what purpose these tools were used:

- DeepL: <https://www.deepl.com/de/translator>, for translations DE <-> EN
- ChatGPT: <https://chatgpt.com/> like a search engine and for GitHub & LaTeX support
- Gwen 3.8 (local installation): proofreading (spelling check and stylistic revision)
- GitHub Copilot in VS Code: support for Python coding

Eigenständigkeitserklärung

Ich versichere, dass ich die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe. Alle Stellen der Arbeit, die dem Wortlaut oder dem Sinn nach anderen Werken entnommen sind, habe ich in jedem einzelnen Fall unter Angabe der Quelle kenntlich gemacht. Dies gilt auch für verwendete Zeichnungen, Skizzen, Ton- und Videoaufnahmen sowie graphische Darstellungen. Ich erkläre mich damit einverstanden, dass meine Arbeit im Verdachtsfall mithilfe einer Plagiatsoftware überprüft wird.

Düsseldorf, October 5, 2026

Marie Vangerow-Kühn